Grade 11 · Algebra II · Topic 6: Exponential and Logarithmic Functions · Lesson 6-6 — Exponential and Logarithmic Equations and Inequalities

Classwork — Give it a go

Show every step. Write which method you use (match the bases, or take a logarithm). Check each logarithm answer against the domain.

Student name :	Date : .....

QUESTION 1 match the bases

1	$4^{x-1} = 64$	Solve for x. Write 64 as a power of 4 first.

QUESTION 2 different bases, one common base

2	$8^{x} = 16^{x-1}$	Solve for x. Write both sides as powers of 2, then check your answer.

QUESTION 3 negative exponent

3	$\left(\frac{1}{5}\right)^{x} = 125^{x+2}$	Solve for x. Hint: write one fifth as a power of 5.

QUESTION 4 exponential inequality, base above 1

4	$5^{2x-3} < 125$	Solve the inequality. State why the direction does or does not change.

QUESTION 5 exponential inequality, base below 1

5	$\left(\frac{2}{3}\right)^{x} \geq \frac{4}{9}$	Solve the inequality. Test one value inside and one outside your answer.

QUESTION 6 logarithmic equation

6	$\log_3(2x-1) = 2$	Solve for x. Convert to exponential form, then check that the argument is positive.

QUESTION 7 logarithmic equation with an extraneous solution

7	$\log_2 x + \log_2(x-7) = 3$	Solve for x. State the domain first, and name any rejected value.

QUESTION 8 logarithmic inequality

8	$\log_5(2x-4) < 1$	Solve the inequality. Write two conditions: the domain, and the inequality itself.

QUESTION 9 use logarithms when the bases will not match

9	$3^x = 17$	Give the exact answer as a logarithm, then to 2 decimal places.

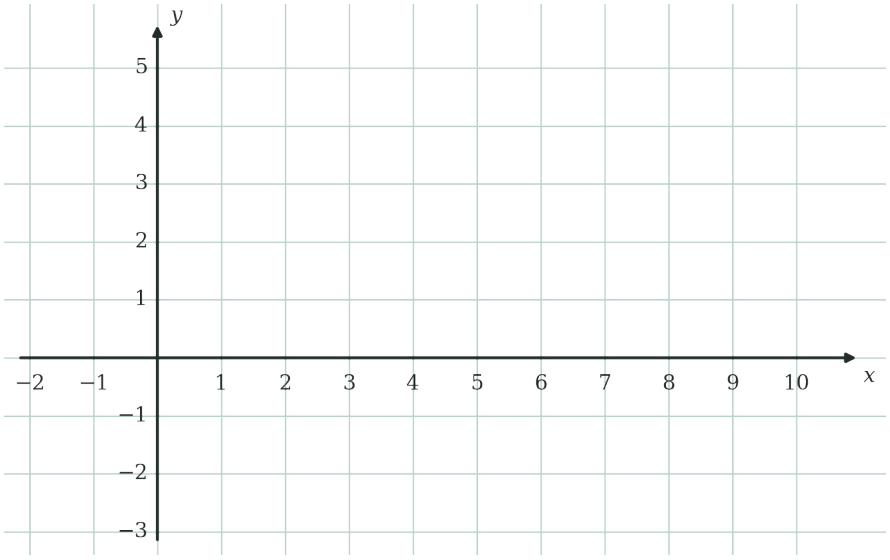
QUESTION 10 model (model values only)

10	$5000\,(1.06)^t = 8000$	A model: 5 000 SAR grows by 6% a year. Find t in years, exact form first, then to 1 decimal place.

QUESTION 11 sketch, then solve using the graph

11	$y = \log_2(x - 1)$	Sketch the graph on the grid below: draw the asymptote as a dashed line and plot the points whose x-values are 2, 3, 5 and 9.

QUESTION 11 (continued) your grid



Use your sketch to solve $\log_2(x - 1) < 2$. Shade the solution on the x-axis and write it below, then confirm it by algebra.